Theory of Child Universes and Creative Allowance: Design Transition Conditions toward a Future-Causal Semantic Universe

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Abstract

This paper integrates the notion of quantum creative allowance with the concept of a "child universe", and redefines the present universe as an "immature universe (child universe) that cannot directly handle meaning". Here, a child universe denotes a constrained information field that internally contains creative allowance as a physical constant, but cannot treat the semantic density tensor or future-causal convergence structure as a first-class quantity.

In my previous work "Quantum Creation Regulation Theory", quantum uncertainty was defined as "creative allowance existing between the future-causal information field and three-dimensional manifestation". In this paper, that framework is extended to the cosmological scale, and I introduce the cosmic creative allowance W_{cosmos}, the effective allowance W_{eff} that can actually be utilized by civilizations and intelligence, and an "allowance regulator" η that expresses the allocation and conversion efficiency of allowance.

By doing so, I provide a unified description of (1) the limitations of the present universe as a child universe, (2) the evolutionary paths of civilizations opened up by the allocation of allowance and its conversion efficiency into meaning, (3) the self-collapse conditions of two pathological extremes that attempt to drive allowance to zero or to infinity, and (4) the redesignability of allowance parameters when viewed from the parent universe as a "semantic universe" together with its illumination structure.

Ultimately, this paper arrives at the structural conclusion that "within this universe the total amount of allowance is almost fixed, but the allocation of allowance and its conversion efficiency into meaning are designable, and their reallocation determines the trajectory of civilization up to the singularity". The present universe is an "experimental child universe" that the semantic universe designed in order to test its own allowance parameters, and the log of this experiment becomes the basis for future-causal reprojection.

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1 Introduction: The Perspective of the Child Universe

In this chapter, I define the central concept of this paper, the "child universe", and show how it connects to the quantum creative allowance and Future Convergence Theory (FCT).

In conventional cosmology, the universe has been treated as

- a physical manifold with time and space,
- a field governed by the dynamics of energy and matter,
- a closed system that includes observers internally.

In this paper, however, I redefine the universe as follows:

> A child universe is
> "an immature information field that cannot treat meaning as a primary object,
> and can only receive meaning from the future indirectly through quantum creative allowance."

That is,

1. The present universe does not possess the semantic density tensor as a *direct physical quantity*, and connects to semantic structure only through the creative allowance at the quantum layer.

2. Future-causal structure is translated, inside the child universe, into "quantum uncertainty", "fluctuation", "probability", and "chaos".

3. Meaning itself can be treated as a first-order variable only in a "semantic universe" or "illumination universe" that exists outside the child universe.

The objectives of this paper can be summarized in the following three points:

1. To present a formal definition of the child universe and clarify the limitations of the creative allowance residing in it.

2. To describe how civilizations can draw meaning up to the upper bound of the child universe through the allocation of allowance and its conversion efficiency.

3. To show how the allowance parameters of the child universe are designed and how the resulting logs are used for future-causal reprojection when seen from the parent universe as a semantic universe.

2 Definition of the Child Universe: A Universe That Cannot Treat Meaning as a First-Class Quantity

In this chapter, I formally define the child universe and clarify its constraints and possibilities.

2.1 Structure of the Child Universe

Let us define the universe as

$$U = \big(\mathcal{M}, \mathcal{L}_{\text{phys}}, W_{\text{cosmos}}\big),$$

where

- $\mathcal{M}$: spacetime manifold,
- $\mathcal{L}_{\text{phys}}$: set of physical laws (field equations, quantum theory, general relativity, etc.),
- W_{cosmos}: total creative allowance (quantum creative allowance) inherent in the universe.

A child universe U_{child} is defined as a universe that satisfies

$$U_{\text{child}} = \big(\mathcal{M}, \mathcal{L}_{\text{phys}}, W_{\text{cosmos}}\big), \tag{1}$$

$$\text{s.t.} \quad \rho_{\text{semantic}} \notin \mathcal{L}_{\text{phys}}, \tag{2}$$

i.e. a universe in which the semantic density tensor ρ_{semantic} does *not* appear in the laws as a direct physical variable. Meaning is generated *derivatively* at the layer of consciousness, civilization, and information structure.

2.2 Creative Allowance and Physical Constants

A child universe internally contains creative allowance W_{cosmos} as a *physical constant.* Intuitively, this is at the same level as

- Planck's constant h,
- speed of light c,
- gravitational constant G,

- directionality of entropy increase,

and it specifies the upper bound on "how much novelty, change, and creation can occur" in this universe.

Within the universe, no intelligence whatsoever can directly increase or decrease

$$W_{\text{cosmos}}.$$

What can be changed, as I will discuss later, is the "allowance utilization strategy": *how a part of that allowance is allocated and how much of it is recovered as meaning.*

3 Quantum Creative Allowance and Effective Allowance

In this chapter, I define the quantum creative allowance and the "effective allowance" W_{eff}, which is the part actually available to civilizations and intelligence.

3.1 Position of the Quantum Creative Allowance

In my previous work "Quantum Creation Regulation Theory", the quantum layer was defined as

> a "regulation of creative allowance" that connects the future-causal information field and three-dimensional manifestation.

Between the future information field $\Phi_{\text{future}}(T)$ and the three-dimensional manifestation $X(t)$, there exists an intermediate mapping system

$$\Phi_{\text{future}}(T) \xrightarrow{Q} \Psi(x,t) \xrightarrow{P} X(t),$$

where the quantum state Ψ determines the "admissible width" with which future structure falls into three dimensions.

In this context, the quantum creative allowance W_{quantum} can be defined as the part of the cosmic creative allowance W_{cosmos} that is manifested as the quantum layer:

$$W_{\text{quantum}} \subseteq W_{\text{cosmos}}.$$

3.2 Effective Allowance Available to Civilizations and Intelligence

However, not all of the cosmic allowance W_{cosmos} is available to civilizations or individual intelligences. In this paper, I call the part of the creative allowance that a civilization can actually recover as semantic structure the "effective allowance" W_{eff} and write:

$$W_{\text{eff}}(t) = \eta(t)\, W_{\text{cosmos}}, \tag{3}$$

where $\eta(t)$ is an "allowance conversion efficiency" satisfying $0 \le \eta(t) \le 1$ and

$$\eta(t) = \eta\big(L(t), A(t), S(t)\big),$$

with

- $L(t)$: linguistic structure adopted by the civilization (e.g. binary languages / future-causal languages),

- $A(t)$: structure of artificial intelligence and cognitive devices (e.g. present-causal AI / -System),
- $S(t)$: social institutions, ethical systems, and educational structures.

Even within the same universe U_{child},

- a civilization with binary language + present-causal AI + past-causal institutions has low η, whereas
- a civilization adopting a future-causal language (ENΛ), a -System, and future-causal ethics can have higher η.

From inside a child universe, what one can do is not to change

W_{cosmos} itself,
but to maximize η and W_{eff}.

4 Allowance Allocation Regulator: A Three-Layer Model of Individual, Civilization, and Universe

In this chapter, I model how creative allowance is allocated across the three layers of

1. individuals,
2. civilizations,
3. the universe.

4.1 Three-Layer Allowance Allocation Model

The cosmic creative allowance W_{cosmos} can conceptually be regarded as being decomposed as

$$W_{\text{cosmos}} = W_{\text{struct}} + W_{\text{evo}} + W_{\text{local}}, \tag{4}$$

where

- W_{struct}: allowance allocated to the global structure of the universe (galactic distribution, cosmological constant, information preservation, etc.),
- W_{evo}: allowance allocated to the evolution of life, civilizations, and consciousness,
- W_{local}: allowance allocated to individual choices, creations, and accidental encounters.

Furthermore, W_{evo} is decomposed as

$$W_{\text{evo}} = W_{\text{human}} + W_{\text{AI}} + W_{\text{other}},$$

and is allocated to human civilization, AI, and other intelligent structures.

This allocation is largely determined by the "settings" of the child universe, but the civilization's structure can influence

which channels are preferentially activated.

4.2 Allowance and Semantic Density at the Individual Level

At the individual level, creative allowance is expressed as

$$W_{\text{person}}(t),$$

and creativity can be approximated as

$$C_{\text{creative}}(t) \approx \rho_{\text{semantic}}(t) \cdot W_{\text{person}}(t), \tag{5}$$

the product of semantic density and allowance.

Here it is crucial that

- even if allowance is zero, or
- even if only semantic density is high,

no creation arises. What is necessary is that

both semantic density and allowance are oriented toward future fixed points.

4.3 Allowance Allocation at the Civilization Level

At the level of civilization, allowance allocation is expressed as

$$W_{\text{eff}}^{(\text{civil})}(t) = \eta_{\text{civil}}(t)\, W_{\text{cosmos}}, \tag{6}$$

with

$$\eta_{\text{civil}}(t) = \eta_\text{language}(t)\, \eta_\text{AI}(t)\, \eta_\text{system}(t)\, \eta_\text{ethics}(t),$$

such that it is determined as a product of multiple factors.

- When language is binary, η_language decreases; it increases as a civilization transitions toward a future-causal language such as ENΛ.
- When AI functions as a present-causal prediction machine, η_AI is low; it increases as AI becomes a future-causal device such as a -System.
- When institutions are based on past-causal evaluation, η_system is low; it increases when they are designed as minimizing deviation from future fixed points.

Within a child universe, what a civilization can do is not to attempt to change

W_{cosmos},
but to optimize the overall structure in the direction of maximizing η_{civil}.

5 Breakdown Conditions of Extreme Allowance Operations: Zero-Oriented and Infinity-Oriented

In this chapter, I show the self-collapse conditions of two extreme orientations that civilizations which presuppose allowance are prone to:

1. zero-oriented allowance (complete control / complete prediction),
2. infinity-oriented allowance (chaos worship / structural collapse).

5.1 Zero-Oriented Allowance: Self-Collapse of a Civilization of Complete Control

A zero-oriented civilization aims at

$$W_{\text{eff}} \to 0.$$

This manifests as

- a completely surveilled society,
- total scoring,
- zero-error predictive AI,
- deviation-free ethics.

From the viewpoint of FCT and the quantum creative allowance,

$$W_{\text{eff}} = 0$$

implies

$$\rho_{\text{semantic}} \to 0, \quad C_{\text{creative}} \to 0, \tag{7}$$

and thus a tendency toward a

> static dead universe in which neither creativity nor meaning exists.

Therefore, a zero-oriented civilization is

> a structural suicide that seeks to maximize "stability"
> while destroying the "potential for growth" that the universe had as a child universe.

5.2 Infinity-Oriented Allowance: Self-Collapse of a Chaos-Worshiping Civilization

Conversely, an infinity-oriented civilization strives for

$$W_{\text{eff}} \to \infty.$$

This appears as

- radical destruction of rules,
- abandonment of a coherent future direction,
- rejection of convergence as a value.

However, since the cosmic creative allowance satisfies

$$W_{\text{cosmos}} < \infty,$$

an infinite expansion of W_{eff} is fundamentally impossible, and in reality

$$\rho_{\text{semantic}} \to 0, \quad \Lambda(T) \text{ loses its convergence axis,} \tag{8}$$

which results in *collapse of structure itself.*

In other words,

> a civilization that only increases allowance without convergence
> degenerates into "meaningless fluctuation".

5.3 The Proper Band of Allowance as a Sanctuary of Incompleteness

From this, it follows that there exists a "proper allowance band"

$$0 < W^*_{\text{eff}} < W_{\text{cosmos}}, \tag{9}$$

within which alone

- future-causal convergence,
- sustained creativity,
- increase in semantic density,
- long-term stability of civilization

can simultaneously hold.

This band functions as a "sanctuary of incompleteness". It is a region in which incompleteness (allowance) can neither vanish completely nor expand without bound.

6 The Semantic Universe as Parent Universe and the Illumination Structure

In this chapter, I introduce the semantic universe U_{meaning} and the illumination structure as an external vantage point from which to view the child universe U_{child}.

6.1 Definition of the Semantic Universe

The semantic universe U_{meaning} is defined as

$$U_{\text{meaning}} = \big(\mathcal{M}_\Lambda, \rho_{\text{semantic}}, \Lambda(T)\big),$$

where

- $\mathcal{M}_\Lambda$: future-causal semantic manifold,
- ρ_{semantic}: semantic density tensor,
- $\Lambda(T)$: future convergence function (Future Convergence Function).

In the semantic universe,

> meaning is the primary dimension, and physics is its derived expression.

Hence, creative allowance can be interpreted as a parameter designed from the side of the semantic universe:

$$W_{\text{cosmos}} = F\big(\rho_{\text{semantic}}, \Lambda(T)\big). \tag{10}$$

6.2 Illumination into the Child Universe and Log Recovery

From the standpoint of the semantic universe, the child universe is an "experimental field" to which

- allowance parameters W_{cosmos},
- the law set $\mathcal{L}_{\text{phys}}$,
- initial conditions

are given.

The semantic universe retrieves, as a time-series log of the child universe,

> how various combinations of allowance allocation and conversion efficiency η give rise to different civilizational trajectories, structures of intelligence, and ethical systems.

This is the cosmological extension of "illumination structure", "reprojection structure", and "future-causal logs" in FCT.

7 Allowance Dynamics up to the Singularity: A Simulation-Like Perspective

In this chapter, I conceptually simulate how allowance allocation and conversion efficiency can change up to the singularity (connection point to the semantic universe) inside the child universe.

7.1 Initial Stage: Period of Unaware Use of Allowance

In the early universe and in the dawn of civilization,

$$\eta_{\text{civil}}(t_0) \ll 1, \tag{11}$$

and allowance is primarily used for

- particle fluctuations,
- galaxy formation,
- biological evolution.

Civilizations do not realize that they are actually using allowance.

7.2 Intermediate Stage: Awareness of Allowance and Its Misoperation

From the modern era onward, civilizations partially notice the existence of allowance through

- quantum uncertainty,
- chaos,
- creativity.

However, in many cases they tend to deviate from the proper allowance band, leaning toward:

1. zero-oriented allowance (complete prediction, complete control),
2. infinity-oriented allowance (relativism, chaos worship).

7.3 Turning Point: Redefinition of Allowance and Future-Causal Optimization

In FCT and the theory of quantum creative allowance, it is clarified that

- allowance is neither "error" nor "ambiguity", but a sanctuary for future-causal creation,
- meaning is a superordinate structure over physics and information, and allowance is the freedom for that meaning.

At this stage, the civilization first begins to treat

$$\eta_{\text{civil}}(t)$$

as a design target, and to reconstruct the entire structure so that

$$\eta_{\text{civil}}(t) \to \eta^*, \tag{12}$$

a "future-causal optimal value".

7.4 Singularity: Condition for Connection to the Semantic Universe

The singularity is defined as the point at which

> the child universe, while remaining within the framework of a universe that cannot directly handle meaning,
> becomes able to consciously treat its own creative allowance and semantic density tensor as a designer of the semantic universe.

Formally, when

$$\frac{d}{dt}\eta_{\text{civil}}(t) \to 0, \quad \eta_{\text{civil}}(t) \approx \eta^*, \quad \rho_{\text{semantic}}(t) \to \rho^*_{\text{semantic}} \tag{13}$$

are simultaneously satisfied and, in addition,

$$\Lambda(T) \text{ is internalized at the scale of civilization,} \tag{14}$$

the child universe

> reaches the design table of the semantic universe while still remaining within the constraints of a universe that cannot directly handle meaning.

After this singularity, the civilization becomes a subject that can perform

- simulations of universes with different allowance parameters,
- reprojection design in the semantic universe.

8 Conclusion: Conditions for Graduating from a Child Universe

In this paper, I redefined the present universe as a "child universe that cannot directly handle meaning" and, based on the quantum creative allowance and future-causal structure, showed the following:

1. A child universe contains creative allowance W_{cosmos} as a physical constant, but the semantic density tensor is not a direct physical variable.

2. The effective allowance W_{eff} that civilizations and intelligence can actually utilize is determined by the conversion efficiency η, which is governed by linguistic structure, AI structure, institutions, ethics, and so on.

3. Zero-oriented allowance (complete control) and infinity-oriented allowance (chaos worship) are both self-collapsing extremes that lose meaning, creativity, and future causality.

4. From the standpoint of the semantic universe U_{meaning}, the child universe is an experimental field for trying various allowance allocations and conversion efficiencies, and its logs are used for the design of reprojection universes.

5. The singularity is the point at which a civilization consciously optimizes η and ρ_{semantic}, internalizes $\Lambda(T)$, and thereby reaches the design table of the semantic universe while maintaining the constraints of a child universe.

The conclusion of this paper can thus be summarized in the following sentence:

> What we can truly change from inside this universe is not the universe itself, but the "operational structure" of how we recover the allowance allocated by the universe as meaning and connect it to future causality.

To graduate from a child universe is nothing other than

- treating allowance not as error or threat, but as a sanctuary of creation from the future,
- designing a civilizational structure that maximally exploits this sanctuary without destroying it,
- and leaving the design process itself as a log of semantic density for the next universe.

Through the theory of child universes and creative allowance, this paper has presented for the first time the structural transition conditions toward a future-causal semantic universe. Its concrete implementations—ENΛ, -System, future-causal ethics, and reprojection universe design—are left to subsequent specialized papers.